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Environmental Taxes and the Choice of Green Technology

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We study several important aspects of using environmental taxes to motivate the choice of innovative and "green" emissions-reducing technologies as well as the role of fixed cost subsidies and consumer rebates in this process. In our model, a profit-maximizing monopolistic firm facing price-dependent demand selects emissions control technology, production quantity, and price in response to the tax, subsidy, and rebate levels set by the regulator. The available technologies vary in environmental efficiency as well as in the fixed and variable costs. Both the optimal policy for the firm and the social-welfare maximizing policy for the regulator are analyzed. We find that the firm's reaction to an increase in taxes may be non-monotone: while an initial increase in taxes may motivate a switch to a greener technology, further tax increases may motivate a reverse switch. For the regulator, we compare the social welfare achievable in the centralized system (which serves as an upper bound) to the highest level achievable under different classes of environmental policies. If the regulator is limited to a tax-only policy, then when the regulator is moderately concerned with environmental impacts, the tax level that maximizes social welfare simultaneously motivates the choice of clean technology and closes the gap to the upper bound; however, both low and high levels of societal environmental concerns may lead to the choice of dirty technology and significant welfare losses as compared to the centralized case. Supplementing the environmental taxation with fixed cost subsidies and consumer rebates can eliminate this effect, expanding the range of parameters over which the green technology is chosen and often closing the welfare gap to the centralized solution.

Key words: green technologies; sustainability; environmental policy; environmental taxes; subsidies and rebates
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1. Introduction

The issues related to the environment and the adoption of "green" technologies have received a large amount of attention over the last several years. Environmentalists won the 2007 Nobel Peace Prize, governments around the world "placed global warming and greenhouse-gas reduction as one of highest priorities" (Pelosi, 2007), and business executives put the issues related to the environment at the top of their agendas. According to a McKinsey survey,[1] 41% of senior executives in the United States and 53% in Europe believe that the environmental issues will have a large impact on shareholder value over the next 5 years. Many industries have developed a number of alternative pollution-abatement technologies, most of them requiring substantial up-front capital investments and, frequently, leading to an increase in variable production costs as well. As a result, the investment in such technologies can often not be justified on purely economic grounds given the prevailing prices. Hence, there is a need for the regulators to use the tools available to them to induce the choice of green technologies. Such tools include environmental taxes, fines, subsidies, consumer rebates, and a variety of other mechanisms (cap-and-trade mechanisms, etc.).

A good example is provided by the cement industry. As outlined in the WBCSD (2009) report, carbon capture and storage (CCS) is the key technology in driving significant carbon emissions reduction in cement manufacturing. The report estimates the cost in 2030 of a new CCS installation/retrofit for an average-size plant to be \$125–\$375 million, with an additional \$12–\$63 million increase in the operational cost per tonne; the useful life of such installation is projected to be 30–50 years. Carbon emission with CCS are projected to be 0.54 tonnes per tonne of clinker, as compared with 2.34 (gas)—4.4 (coal) tonnes without. The report then discusses a variety of regulatory actions needed to motivate the industry to adopt CCS. In particular it states that "a long term global

carbon price of \$50–\$100 per tonne of CO_2, rising to \$200/tonne by 2050 is likely to be needed to provide returns on CCS investments"—with the non-CCS emission levels quoted above, this implies that the cost of CO_2 emissions alone should be in the over \$100 per tonne of cement. Given the current market price of cement of about \$100 per tonne, firms are clearly unlikely to invest in CCS on their own. Thus the report mentions several possible policy mechanisms, including carbon taxes and a variety of subsidies: for installation, via earmarked government loans, for research and development, etc.

The taxation approach recently gained significant traction among regulators (e.g., Hargreaves 2010) and in the business media (e.g., it is strongly supported by *The Economist*), primarily because it provides a direct measure of the price of emissions to the firm. A classic example of the successful use of environmental taxation is the US tax on chlorofluorocarbons (CFCs) introduced on January 1, 1990, following the 1987 adoption of the so-called Montreal Protocol aimed at eliminating the manufacture, trade, and use of ozone-destroying substances. At inception the tax was already higher than the cost of CFCs themselves, and over the following 5 years the tax doubled. As a result of the US tax and measures taken by other countries, by 1995 the global consumption of CFCs had dropped by over 70%. We note that the primary use of CFCs was in refrigeration, air conditioning, and foam insulation, and there is no evidence of a reduction in the use of any of these products. Thus, the primary effect of the environmental taxation has been to induce the choice of less environmentally damaging technology alternatives (it should be noted that such alternatives were available when the tax was introduced). Interestingly, the use of CFCs dropped by only a few percent between 1987 and 1990 before the US tax was introduced, supporting the effectiveness of environmental taxation as a policy tool; see Oberthür (1997) for further details.

Environmental taxation (unlike, say, prohibitions on the use of certain technologies) is an indirect tool, with the regulators trying to provide proper incentives for firms to make the "right" technology choices. The goal of the current study is to explore the possibilities and limitations of this tool, together with related tools such as subsidies and rebates. The questions we ask are: Under what conditions can a certain technology choice be induced? What are the economic impacts of these tools, including impacts on consumer prices, production quantities, profitability of the firms, pollution quantities, etc.? To this end, we consider a leader–follower Stackelberg game between two key players: a profit-maximizing firm (the follower) and a social welfare-maximizing regulator (the leader).

The first player is the profit-maximizing firm that produces some desirable good (e.g., cement, refrigerators, etc.) and, as a natural by-product of its production process, emits an undesirable pollutant (e.g., carbon dioxide, CFC, etc.) The firm has a number of alternative pollution reduction technologies available,[2] differing in the environmental efficiency as well as in fixed and variable costs. The firm faces a price-sensitive consumer demand, as well as the environmental policy set by the regulator, consisting of taxes, subsidies, and consumer rebates. The technology choice affects the amount of pollutant emitted (and thus the total environmental tax), as well as the fixed and variable production costs. Certain "green" technology choices may also be eligible for subsidies and rebates.

Our setting implicitly assumes that the customer demand is not affected by the choice of the technology. This is a reasonable assumption in case of so-called "end-of-pipe" pollution control technologies, where the technology choice affects only the amount of the pollutant, but not the characteristics of the final product—as is the case in the cement and CFC examples above. While it can be argued that consumers may be willing to pay extra for a good produced with a greener technology, findings from an OECD (2011) report indicate that the number of such consumers is quite low. The situation may be quite different when the green technology modifies the product, resulting in higher demand from the so-called "green consumer segment"—consumers who are willing to pay extra for greener alternatives (e.g., a popular hybrid-electric car, the Toyota Prius, is sold at a premium as compared to non-hybrid alternatives). Our study considers the former case: the firm is not changing its product/service, but produces it with a different/greener technology. Having said that, later in the article we indicate how our modeling framework can be extended to account for an increase in demand for products produced with green technology.

The second player is the regulator, who, acting as a Stackelberg leader, has a number of environmental policy tools at its disposal: the environmental tax assessed per unit of pollutant emitted into the environment, fixed and variable cost subsidies given to the firm if it chooses a particular green technology, and rebates extended to consumers if they buy products manufactured with green technologies. The regulator pursues two goals: maximize the "economic surplus" consisting of firm's profit and consumers' surplus, and minimize the amount of emissions. As we show, these two goals are necessarily conflicting: the economic surplus is maximized by setting the environmental tax to zero, in which case the profit-maximizing choice for the firm is the technology leading to the largest possible emissions. Thus we

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introduce a weighted objective for the regulator, which we call the *social welfare* function, consisting of the sum of economic surplus and the total emissions multiplied by the "environmental concern" parameter ε. This parameter plays an important role, as it essentially converts a unit of emissions (e.g., a ton) into a monetary unit of economic surplus (a $), measuring the degree of society's environmental concerns with respect to the pollutant emitted by the firm or its physical hazardness.

In our analysis we first solve for the firm's (follower's) optimal response to the regulator's environmental policy, and then embed this response into the regulator's (leader's) problem to derive the optimal environmental policy. We benchmark the resulting optimal value of the social welfare function against the so-called "centralized" (first-best, coordinated) solution—a theoretical upper bound obtained by allowing the regulator to have direct control over the firm's decisions. The gap between the value in the Stackelberg game and the centralized solution can be used as a measure of the effectiveness of the environmental policy. Our analysis generates three sets of insights:

Insights for the firm. Our core result is that, contrary to the "conventional wisdom" (as expressed, e.g., in Ramseur and Parker 2009) increasing the environmental tax does *not* necessarily motivate the firm to choose a greener technology. The firm's reaction to taxation may, in fact result in the *non-monotone technology choice*: sufficiently high tax rates may induce the choice of dirtier rather than cleaner technologies. In addition, we show that the case with three or more technologies has fundamental differences from the two-technologies case: the regulator may not be able to motivate the choice of some desirable technology through taxation, even if that technology can be induced in a pair-wise comparison with any other technology.

Insights for the regulator in setting the optimal tax level. From the technical perspective, we show that although the regulator's objective is highly non-linear and discontinuous, the set of potentially optimal tax levels is small and includes certain locally optimal points (which we characterize in closed form), as well as the breakpoints at which the firm changes technology. From the managerial perspective, the optimal tax level and the technology choice it induces naturally depend on the value of the weight parameter representing the degree of the society's environmental concerns. For the cases with a medium level of environmental concern, the socially optimal tax level induces the choice of the green technology, and in many of these cases the regulator is able to achieve the centralized solution. However, when the environmental concern parameter is either very low or very high, the social welfare may be optimized by selecting the tax rate that induces the choice of dirtier technologies—this, again, contradicts the conventional wisdom. We also show that for a broad range of values of the environmental concern parameter, the optimal value of tax, the technology choice it induces, and the resulting production quantity and price are insensitive to the exact value of the parameter. This finding is important because the value of this parameter may be hard to estimate precisely in practice.

Insights for combining taxation and subsidies/rebates mechanisms. While the effects of variable cost subsidies are shown to be equivalent to the environmental taxes, both fixed cost subsidies and consumer rebates are shown to be essential components of the effective environmental policy. They play complementary roles. Rebates are efficient in increasing the welfare for a given technology choice to that of the centralized solution, but may not be able to ensure that the firm selects that technology. Subsidies, on the other hand, are efficient in ensuring that the firm selects a desirable technology without impacting the welfare obtained with that technology. Therefore, with the combined environmental policy that uses taxes, subsidies, and rebates, the regulator can *always* reach the centralized (i.e., highest possible) welfare when the green technology is optimal in the centralized solution. The only case when combined policy is unable to reach the centralized welfare is when the dirty technology is optimal in the centralized solution, which happens when the society's environmental concerns are very low.

In the remainder of the article, section 2 discuss the relevant literature. Sections 3 and 4 present the models of the firm and the regulator. Section 5 discuss the optimal environmental policy and the role of taxes, subsidies, and rebates. Section 6 summarizes and discusses future research. An Online Appendix presents the proofs and some model extensions.

We close the introduction by noting that in this study we consider the case of a monopoly, which has some similarity with Bertrand oligopoly (intuitively, the firm with the technological advantage can undercut all other firms and become a monopolist for the range of tax rates over which its technology choice is optimal). The case of a Cournot oligopoly, where all firms charge the same price, is much more complex and is discussed in a follow-up study.

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2. Literature Review

Our study is related to two streams of literature: one in operations management and another in economics. In the operations literature, a number of researchers addressed environmental problems with respect to remanufacturing (e.g., Debo et al. 2005, Ferguson and Toktay 2006, Majumder and Groenevelt 2001, Souza et al. 2007). Several authors address the product take-back legislation (e.g., Atasu et al. 2009, Atasu and Subramanian 2009). These works consider a tax that is charged per unit sold and analyze the incentives this legislation provides for manufacturers to design products to be more recyclable. Our work differs from the above in several dimensions; most importantly, the question we address is broader—we study production and pollution in general, not just recycling/remanufacturing. At the same time, our model is similar to the above in the sense that we too consider a managerially relevant framework that takes into account operational details of firms' technology choice decisions.

In the economics literature, the questions addressed are more similar to ours, but the framework is very different—(see Jaffe et al. 2002, Requate 2006 for recent reviews). First, a rather common assumption in the economics literature is that a firm's cost of reducing (abating) emissions is given by a continuous twice-differentiable function; for example, see the fundamental work of Barnett (1980) or more recent treatment by Requate (2006). This implies that a continuum of technologies is available and the firm is deciding on the cost of abating emissions, aka "technology." In many cases, however, the managerial decision is very different: the firms must decide whether to invest or not in a given technology from a certain set of available technologies—a discrete choice that underlies our modeling framework. Discreetness also differentiates our work from Carraro and Soubeyran (1996), who assume that the firm possesses both "dirty" and "clean" technologies and consider the continuous problem of allocating production between plants with those technologies.

Second, a common assumption is that the setup costs for the new technology are equal to zero (e.g., Montero 2002, Requate 1998). Our assumption of non-zero fixed costs is certainly much more realistic, but also complicates the analysis because the firm's cost and profit becomes non-differentiable and possibly discontinuous. As we discuss later in the article, the firm's response becomes monotone only if all technologies have the same fixed cost. Requate (1998) establishes monotone response, which occurs because fixed costs are not considered. His model is similar to ours in this case; a slight advantage of our result is that it holds for any demand function. Amacher and Malik (2002) consider fixed cost but do not consider revenues and thus also establish monotone response. Note, however, that none of those studies discuss the non-monotone response and its implications.

Third, a typical assumption is to frame a firm's reaction to environmental regulation as a cost minimization problem (e.g., Amacher and Malik 2002, Fischer et al. 2003). In our model the firm sells its product in a market with price-sensitive demand: in practice, managers worry about the effects of increased costs on price and consequently on the quantity demanded. The latter is particularly salient because in order to cover the fixed costs of a given technology the firm may need to have a large enough production quantity.

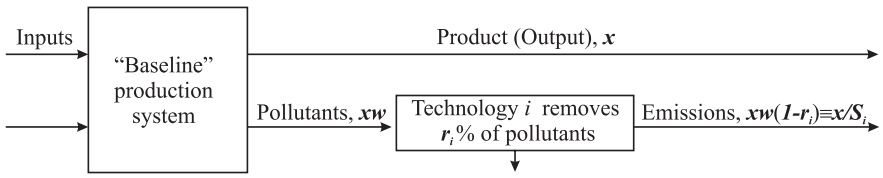
To the best of our knowledge, the only other study that combines the economics and operations approaches and analyzes the technology choice under emissions regulations is the working paper by Drake et al. (2010), who consider both tax and cap-and-trade regulatory regimes in a setting similar to the Carraro and Soubeyran (1996), with all the respective differences from our study; they also do not explicitly consider the regulator's problem.

3. Model of the Firm's Technology Choice

In this section, we consider the model of the firm acting in response to the environmental policy set by the regulator. We initially focus on the firm's response to taxation (for fixed levels of subsidies and rebates), and consider the effects of the other instruments later.

The schematic model of the firm is presented in Figure 1. The firm's "baseline" production system allows it to produce x units of output at the cost $K + cx$, where $K, c \geq 0$ represent "baseline" fixed and variable production costs. As a byproduct of the production process, the firm produces w units of pollutant per unit of output—for the total amount of xw. The

Figure 1 Model Schematic



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amount of the pollutant can be reduced by employing emission-reducing technology. The firm has a choice of a finite number of such technologies, numbered 1,…,k. Technology i removes $r_i(\times 100\%)$ of pollutants, so that the amount emitted equals $xw(1 - r_i)$. To simplify the notation, we let $S_i = \frac{1}{(1-r_i)w} > 0$, and thus x units of output lead to the emission of x/S_i units of pollutant. Note that the higher the value of S_i, the more effective the technology: for example, if $S_2 = 2S_1$, technology 2 reduces the amount of pollution by half vs. technology 1. We assume that technologies are ordered by their effectiveness, that is, $S_1 < S_i < S_{i+1}, i = 2, \ldots, k - 1$, where technology 1 is the "baseline" or "null" technology.

Each technology impacts the fixed and variable production costs. Specifically, technology i is described by two cost parameters: Θ_i, which denotes the one-time[3] fixed (installation, etc.) cost, and ξ_i, which denotes its variable (operating, etc.) cost per unit of product. Thus, if technology i is selected, the fixed and variable costs become $K + \Theta_i$ and $c + \xi_i$, respectively. For technology 1 the costs are already incorporated into the baseline production costs; hence $\Theta_1 = \xi_1 = 0$.

Observe that for every technology i either $\Theta_{i+1} > \Theta_i$ or $\xi_{i+1} > \xi_i$ (or both) must hold—otherwise technology i is less effective and more expensive than technology $i + 1$ and can thus be dropped from consideration. Note that our description of technologies is very general; we do not assume any functional relationship between effectiveness (S_i) and costs (Θ_i, ξ_i).

The regulator imposes tax $t \geq 0$ per unit of pollutant emitted. Thus, if the firm produces x units of output and selects technology i, the total tax charge it faces is $\frac{tx}{S_i}$. This leads to the following production cost function:

$$C_i(x, t) = (K + \Theta_i)I_{x>0} + (c + \xi_i + t/S_i)x, \qquad (1)$$

where I_Φ is the indicator function for set Φ. Note that the firm always has a choice to shut down, incurring no production costs.

Let $D(p)$ be the demand function faced by the firm, where p is the price offered to consumers.[4] Note that at equilibrium $x = D(p)$, allowing us to use the decision variables x and p interchangeably. For a given tax level t, technology choice i, and price p, the firm's profit is given by

$$Pr_i(p, t) = pD(p) - C_i(D(p), t). \qquad (2)$$

The firm's best response strategy to a given tax level t can be viewed as a two-stage problem. First, for each technology choice i the firm optimizes its market response $p_i^*(t)$ by solving $Pr_i(t) = \max_{p \geq 0} Pr_i(p, t)$. Second, the firm optimizes its technology choice $i^*(t) = \arg\max_{i \in \{1,\ldots,k\}} Pr_i(t)$. The firm's strategy is thus described by the pair $\{i^*(t), p^*_{i^*(t)}(t)\}$, with the optimal production level $x^*_{i^*(t)}(t) = D(p^*_{i^*(t)}(t))$. This can be regarded as the firm's reaction function to the tax level t in the leader–follower Stackelberg game.

Remark 1. Our demand function is independent of the technology choice, implicitly assuming that consumers are not willing to pay more for a product produced with greener technology. As noted in the Introduction, this is a reasonable assumption for the "end-of-pipe" technologies (as depicted in Figure 1) or, more generally, for settings where technology choice does not affect the properties of the product. There are, of course, cases where the technology choice does affect the final product (as in the case of Toyota Prius vehicles cited earlier), or there exists a "green market segment." Our model can be extended to this case, for example, by letting ρ_i represent the relative premium associated with technology i, with $1 = \rho_1 \ldots \leq \rho_k$ and letting the demand associated with technology i be given by $D_i(p) = \rho_i D_1(p)$.

Definition 1. *Technology $i \in \{1,\ldots,k\}$ is said to be inducible if there exists $t \geq 0$ such that $i^*(t) = i$ and $p^*_{i^*(t)}(t) > 0$, that is, the firm's optimal reaction to taxation level t is to select technology i and some positive production quantity.*

While some of the results that follow apply to general demand functions, to simplify the exposition and unless stated otherwise, we will assume a constant elasticity demand function:

$$D(p) = Ap^{-\delta}, \qquad (3)$$

where $\delta > 1$ is the elasticity of demand and $A > 0$ is the scaling constant, denoting, for example, market size. All our results hold for the linear demand function $D(p) = a - bp$ as well; see the Appendix. In view of (1), (2), and (3), for a given technology i and tax level t, and assuming the firm decides to produce, the optimal price and quantity are given by

$$p_i^*(t) = \delta(c + \xi_i + t/S_i)/(\delta - 1), \qquad (4)$$

$$x_i^*(t) = \bar{A}(\delta - 1)(c + \xi_i + t/S_i)^{-\delta}, \qquad (5)$$

where $\bar{A} = A(\delta - 1)^{(\delta-1)}/\delta^\delta$, and the associated optimal profit is

$$Pr_i(t) = \bar{A}\left(c + \xi_i + \frac{t}{S_i}\right)^{1-\delta} - (K + \Theta_i). \qquad (6)$$

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Note that $Pr_i(t)$ is a decreasing convex function of t. It is easy to show that $Pr_i(t) > 0$ iff $0 \le t < t_i^{lim}$, where $t_i^{lim} = S_i\left[\left(\frac{\bar{A}}{K+\Theta_i}\right)^{1/(\delta-1)} - c - \xi_i\right]$. Let $t^{lim} = \min_{i\in\{1,\ldots,k\}} t_i^{lim}$.

We make the following assumptions.

Assumption 1. $t^{lim} > 0$.

Assumption 2. $Pr_1(0) \ge Pr_2(0) \ldots \ge Pr_k(0)$.

The first assumption holds w.l.o.g. and helps us to avoid trivial cases: if $t^{lim} \le 0$ then there is some technology $j \in 1,\ldots,k$ that is never selected, and can therefore be removed from consideration. The second assumption states that for any pair of technologies i,j with $i < j$ (i.e., j is cleaner than i), j is not strictly preferred at the $t = 0$ tax level, since otherwise the taxes are not required to motivate a switch *to* a cleaner technology; in fact taxes can only motivate a switch *away* from it. Thus, from the point of view of our study, this is not an interesting case (in fact, Assumption 2 can be relaxed to only require that $i^*(0) \neq k$ with no significant affect on the results, but at the expense of more cumbersome notation).

As noted earlier, the regulator, in addition to taxes, has several other policy tools. For technology i these include the fixed cost subsidy FC_i, the variable cost subsidy VC_i, and the consumer rebate R_i. To incorporate the subsidies into our model, we simply substitute $\Theta_i(FC) \equiv \Theta_i - FC_i$, $\xi_i(VC) \equiv \xi_i - VC_i$, in place of Θ_i and ξ_i in the expressions above. Likewise, if consumers receive a rebate of R_i percent for purchasing a product produced with technology i, then we redefine $A(R) \equiv A(1 - R_iI_i)^{-\delta}$ where I_i is an indicator that technology i has been selected. With these substitutions, Equations (4)–(6) continue to apply. While it may be more precise to define a firm's optimal response (i^*, p^*) to the strategy $(t, FC_i, VC_i, R_i, i = 1,\ldots,k)$ of the regulator, we will generally suppress all the components other than t until the latter sections where the explicit effect of these components is analyzed.

Remark 2. Note that both the environmental tax t and the variable cost subsidy VC_i affect the variable cost component in the model: assuming technology i is selected, the effective variable cost is $c + \xi_i + t/S_i - VC_i$. It is thus clear that the variable cost subsidy can be grouped together with the environmental tax by suitably redefining tax level t. Thus, to simplify the notation we will not consider the variable cost subsidy as a separate policy tool. We note that this involves only a slight loss of generality: in our analysis we generally require that $t \ge 0$, while the combined tax and variable cost rebate may result in effective tax being negative. While our analysis is easily extendable to that case, we will not consider it further.

3.1. Critical Taxation Levels and Non-monotone Response

In this section we analyze the optimal technological choice made by the firm in response to taxation level t (the subsidy and rebate components are assumed to be fixed). In particular, we seek to answer whether taxes can be used to induce the switch to a cleaner technology. We start by discussing the two-technology case.

Let $\Delta(t)$ be the difference in optimal profits:

$$\Delta(t) = Pr_2(t) - Pr_1(t). \tag{7}$$

Thus, if $\Delta(t) > 0$, then the cleaner technology 2 would be selected at tax level t and vice versa. Note that by Assumption 2, $\Delta(0) < 0$ and thus technology 2 is inducible if and only there exists $t \in [0, t_2^{lim})$ for which $\Delta(t) > 0$.

With respect to costs, recall that, by assumption, either $\Theta_1 < \Theta_2$ or $\xi_1 < \xi_2$ must hold. Moreover, if $\Theta_1 < \Theta_2$ but $\xi_1 \ge \xi_2$, then clearly the cost advantage of technology 2 is greatest when tax $t = 0$. Since $Pr_2(0) < Pr_1(0)$, this implies that technology 2 will not be chosen for any tax level. Thus, the only interesting case from our viewpoint is $\xi_1 < \xi_2$, for which we have the following result (please note that all proofs are in the Appendix).

Proposition 1. *If $\xi_1 < \xi_2$, then $\Delta(t)$ is quasi-concave with a unique maximum at*

$$t^{max} = \min\left[\frac{(c+\xi_2)(S_2/S_1)^{1/\delta} - (c+\xi_1)}{1/S_1 - 1/S_2(S_2/S_1)^{1/\delta}}, t_2^{lim}\right]. \tag{8}$$

That is, the clean technology 2 is inducible if, and only if, $\xi_1 < \xi_2$ and $\Delta(t^{max}) > 0$.

First note that t_2^{lim} is increasing in A, while the first term on the right-hand side in Equation (8) is independent of A. Thus, if $\xi_1 < \xi_2$, technology 2 is inducible whenever the market size A is sufficiently large.

From quasi-concavity, $\Delta(t)$ is increasing for $t \in [0, t^{max}]$ and decreasing for $t \in [t^{max}, t_2^{lim}]$. Since $\Delta(0) < 0$, this implies that when technology 2 is inducible, equation $\Delta(t) = 0$ has at least one and possibly two roots on $[0, t_2^{lim}]$. Let us denote these roots by t_{12}^{crit} and t_{21}^{crit}, where $t_{12}^{crit} \le t_{21}^{crit}$. Observe that at the smaller root t_{12}^{crit}, the sign of $\Delta(t)$ switches from negative to positive, indicating a switch from technology 1 to 2. Similarly, at t_{21}^{crit} the switch from technology 2 back to technology 1 occurs. Because the optimal technology

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choice changes at these taxation levels, we refer to them as *critical*. We note that while $t_{12}^{crit} < t_2^{lim}$ is assured whenever technology 2 is inducible, the existence of $t_{21}^{crit} < t_2^{lim}$ depends on the problem parameters. This discussion is summarized in the following proposition.

PROPOSITION 2. *Assume the values of $\delta > 1$, $K, c > 0$, $0 \le S_1 < S_2$, and $0 \le \xi_1 < \xi_2$ are given and A is sufficiently large (such that $t_2^{lim} > t^{max}$). Then technology 2 is inducible, and the function $\Delta(t)$ has at least one and at most two roots on $t \in [0, t_2^{lim})$, with $0 < t_{12}^{crit} < t^{max} \le t_{21}^{crit} \le t_2^{lim}$. In general three regions can be identified:*

$$\begin{array}{lll} \textit{Region I}: & 0 < t < t_{12}^{crit} & \textit{Technology 1 (''dirty'') is preferred} \\ \textit{Region II}: & t_{12}^{crit} < t < t_{21}^{crit} & \textit{Technology 2 (''clean'') is preferred} \\ \textit{Region III}: & t_{21}^{crit} < t \le t_1^{lim} & \textit{Technology 1 (''dirty'') is preferred}, \end{array} \tag{9}$$

with the firm indifferent between two technologies for $t \in \{t_{12}^{crit}, t_{21}^{crit}\}$. The necessary condition for the existence of Region III is $\Theta_2 > \Theta_1$. Moreover, Region III is not empty provided Θ_2 is sufficiently large.

This proposition presents our main result for the firm's problem, indicating that the optimal technology choice may be non-monotone with respect to taxation: increasing the tax level does not necessarily favor the cleaner technology. The following example illustrates this non-monotonicity.

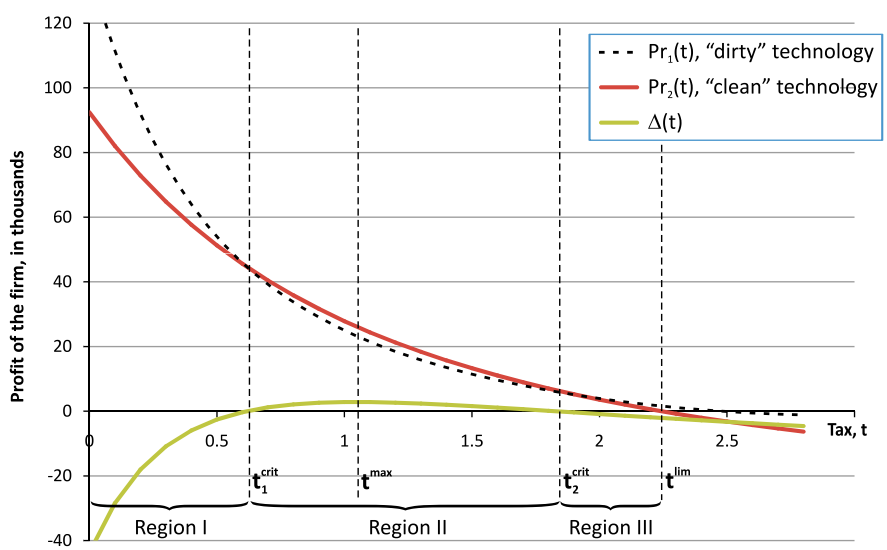
EXAMPLE 1. Suppose the market size $A = 1{,}000{,}000$, demand elasticity $\delta = 3$, and baseline fixed and variable production costs are given by $K = 12{,}000$ and $c = 1$, respectively. Assume technology 1 is null, with $S_1 = 1$, $\xi_1 = \Theta_1 = 0$, and technology 2 has the following parameters: $S_2 = 2$, $\xi_2 = 0.1$, $\Theta_2 = 18000$. That is, the cleaner technology 2 reduces emissions (and tax obligations) by a factor of two, has installation costs of 18,000, and increases variable production costs by 0.1 per unit (i.e., by 10%).

It can be computed (numerically) that $t_2^{lim} = 2.244$, $t_{12}^{crit} = 0.616$, and $t_{21}^{crit} = 1.828$. The resulting functions $Pr_1(t)$, $Pr_2(t)$, and $\Delta(t)$ are illustrated on Figure 2. Here the "dirtier" null technology 1 will be selected for $0 < t < 0.616$ (Region I) and $t > 1.828$ (Region III), while the cleaner technology 2 is only preferred for $0.616 < t < 1.828$ (Region II).

The reason why the *non-monotone technology choice* effect illustrated above may arise at higher taxation levels has to do with the fixed costs. Increasing taxes has two effects: decreasing the variable production cost of technology 2 relative to technology 1 (since $t/S_1 > t/S_2$), but also increasing the firm's total production costs. While the first effect makes technology 2 more attractive as t increases from 0 up to t^{max}, after the level t^{max} is crossed, the second effect takes over. Increase in the total production costs triggers an increase in the optimal market price, which leads to lower optimal demand and hence lower equilibrium production quantity, reducing the total revenue of the firm. The reduced revenue may not be sufficient to offset the higher fixed costs Θ_2 of technology 2, and hence we have non-monotonicity.

The existence of Region III is fairly common. Extending the above example we drew parameter values uniformly from $\delta \in (1,5]$, $S_2 \in (1,5]$, $\xi_2 \in [0,1]$, and $\Theta_2 \in [0, 20{,}000]$ for $S_1 = 1$ and observed that Region III is non-empty in ≈27% of problem instances.

Figure 2 The Existence of Three Critical Regions, Illustration for Example 1



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From the proof of Proposition 2, we can establish the following corollary.

COROLLARY 1. *The value of t_{12}^{crit} is increasing in the fixed cost Θ_2 and the value of t_{21}^{crit} is decreasing in Θ_2. Therefore the size of Region II, given by $t_{21}^{crit} - t_{12}^{crit}$, is decreasing in Θ_2.*

The preceding result says that as the installation cost Θ_2 decreases, Region II expands. This suggests that subsidies given to the firm to (fully or partially) offset the installation cost of the cleaner technology may be effective in expanding the region where the cleaner technology is chosen.

As noted earlier, all of the structural results discussed above hold for the case of linear demand functions; see the Appendix. We also conjecture that the results, in fact, hold for substantially more general demand and productions functions—however, the resulting behavior of the $\Delta(t)$ function may be more complicated. In particular, there may be more than two critical taxation values present (i.e., there may be more than two alternating regions where each technology is preferred).

The results above also extend to the case with three or more technologies, but with some complications. In particular, pair-wise comparison of a given technology against each alternative is not sufficient to establish its inducibility. Region II of one pair can overlap with Region I or III of another, and so a technology can be inducible in any pair-wise comparison, but not inducible overall. Working through the three-plus technologies case requires some additional notation; please see the Appendix for details.

Regardless of the number of technologies, however, the problem is greatly simplified if the fixed costs of all technologies are equal, as we discuss below.

PROPOSITION 3. *Suppose $\Theta_i = \Theta_k$ for all $i = 1,\ldots,k$ and let $D(p)$ be any non-negative continuous non-increasing (i.e., "demand") function. Let $t_{ik} = (\xi_k - \xi_i)/(1/S_i - 1/S_k)$. Then technology k is inducible iff*

$$t_{*k}^{crit} \equiv \max_{\{i=1,\ldots,k-1\}} t_{ik}^{crit} < t_k^{lim}$$

The preceding result ensures that if there is no difference in the fixed costs between the technologies, then the non-monotone technology choice effect disappears. Note that the proposition holds for any demand function. We also note that Assumption 2 ensures that $t_{*k}^{crit} > 0$, and Proposition 3 implies that technology k is inducible whenever it is pairwise-inducible with respect to every other technology. The conditions in Proposition 3 are also easy to check since all quantities can be computed in closed form. The difference between Propositions 2 and 3 shows the importance of incorporating the installation costs of technologies in the analysis.

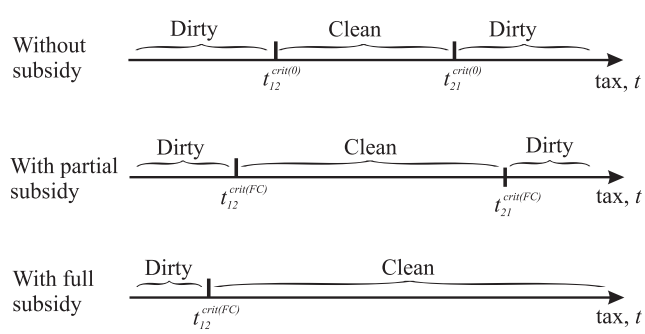
To summarize, two factors determine whether the cleanest technology k is inducible as compared with other available technologies: the installation cost Θ_k compared with the installation costs of other technologies and whether k's environmental efficiency S_k is sufficient to offset the (possibly) higher operating cost ξ_k at some feasible taxation level t. The preceding results suggest that by subsidizing (either fully or partially) the installation costs of the new technology, the regulator can induce the choice of technology k in cases where it would not be chosen otherwise—this is illustrated in Figure 3. In fact, by fully subsidizing the fixed costs, the cleanest technology can be induced whenever it is competitive on the operating-cost-only basis. We will examine the issue of fixed cost subsidies and their effects further in the following sections. We next turn our attention to the regulator's problem.

4. Regulator's Problem: Model of Social Welfare

In this section we analyze the problem of the regulator (Stackelberg leader) that anticipates the firm's reaction to the environmental policy and thus sets the levels of taxes, subsidies, and rebates strategically to maximize the social welfare. We start by defining the social welfare function.

To simplify the exposition, we will restrict ourselves to the two-technology case, where technology 1 is the "dirty" technology and technology 2 is the "clean" one, that is, with $S_2 > S_1$. The regulator's policy tools are the tax level t, the fixed cost FC, and the consumer rebate R—since the subsidy/rebate only applies to cleaner technology, there is no need for the subscripts (by Remark 2, there is no need to consider the variable cost subsidy as a separate policy tool). Thus the regulator's policy can be expressed as (t,FC,R). As before, we let $\Theta_i(FC) = \Theta_i - FC \times I_{i=2}$ and $A(R) = A(1 - R\times I_{i=2})^{-\delta}$, where $I_{i=2}$ is the indicator function for the choice of technology 2. All expres-

Figure 3 The Effect of Subsidizing Fixed Costs on Critical Taxation Levels



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sions in the previous section can now be applied directly by using $\Theta_i(FC)$ and $A(R)$ in place of Θ_i and A, respectively.

The economic system we are considering consists of three stakeholders: the firm that produces the (desired) good and emits the (undesired) pollutant, the consumers who are enjoying surplus from consuming the good and the surplus from the government services but are also adversely affected by the pollutant emitted, and the regulator that sets the tax/subsidy/rebate levels and collects tax revenue, of which a part is spent on subsidy and rebate and remaining net tax revenue is spent on the services that benefit consumers. We thus assume that the social welfare function consists of the following components:

$$\text{Social Welfare} = \text{Firm's Profit} + \text{Consumer Surplus} - \text{Environmental Impact}. \tag{10}$$

Suppose the regulator selects policy (t,FC,R) and the firm's technology choice is $i \in 1,2$. We will generally assume that the regulator's policy is chosen to ensure that the firm will not shut down. Using the expressions derived in the previous section, the components of the social welfare function are given as follows.

Firm's Profit is given by Equation (6) and with $\Theta_i(FC), A(R)$ as defined above is equal to

$$\text{Firm's Profit} = \bar{A}(R)\left(c+\xi_i+\frac{t}{S_i}\right)^{1-\delta} - K - \Theta_i(FC),$$
$$\text{where } \bar{A}(R) = A(R)(\delta-1)^{(\delta-1)}/\delta^{\delta}.$$

Consumer Surplus is composed of two components: surplus from consuming the firm's product and surplus from the government services, where the latter equals the net tax revenue after paying for the cost of subsidies and rebates, which is "returned" by the regulator to consumers in the form of various services or direct refunds. The first component is defined as the area under the demand curve above the optimal price. Integrating the demand function from Equation (3) and substituting the expression for the optimal price from Equation (4) we obtain that surplus from consumption $= \int_{p_i^*(t)}^{\infty} D(w)dw = \frac{1}{\delta-1}A(R)[p_i^*(t)]^{(1-\delta)} = \bar{A}(R)\frac{\delta}{\delta-1}(c+\xi_i+\frac{t}{S_i})^{1-\delta}$.

The second component equals the environmental tax revenue collected from the firm, which equals $tx_i^*(t)/S_i$, minus the cost of of the subsidy, $FC \times I_{i=2}$, minus the total rebate. The latter equals the per unit rebate $I_{i=2} \times Rp_2^*$ times the production quantity $x_2^* = D(p_2^*)$. Substituting the expressions from Equations (4) and (5) and combining all components, we obtain

$$\text{Consumer Surplus} = \bar{A}\delta\left[\frac{1}{\delta-1} - R\times I_{i=2}\right]\left(c+\xi_i+\frac{t}{S_i}\right)^{1-\delta} + t\frac{\bar{A}(\delta-1)}{S_i}\left(c+\xi_i+\frac{t}{S_i}\right)^{-\delta} - FC\times I_{i=2}.$$

Environmental Impact is obtained by multiplying the amount of pollutant emitted by a coefficient $\varepsilon \geq 0$ that translates a firm's emissions into monetary units. Using Equation (5),

$$\begin{aligned}\text{Environmental Impact} &= \varepsilon\frac{x_i^*(t)}{S_i}\\ &= \varepsilon\frac{\bar{A}(R)(\delta-1)}{S_i}(c+\xi_i\\ &\quad + t/S_i)^{-\delta}.\end{aligned}$$

The coefficient ε plays an important role in our model, as it measures the degree of environmental concern/awareness of the regulator and society. It also has a physical meaning, specifying in monetary terms the degree of environmental hazard of the firm's physical production process and the associated pollutant.[5] Higher ε means that the society perceives a more significant welfare loss because of emissions and is thus more interested in reducing the level of pollution.

With these definitions, for a given environmental policy (t,FC,R), technology choice $i \in \{1,2\}$, and assuming $Pr_i(t) \geq 0$ (i.e., no shutdown), the social welfare is given by

$$\begin{aligned}Q_i(t,FC,R) = &(t-\varepsilon)\frac{\bar{A}(R)(\delta-1)}{S_i}(c+\xi_i+t/S_i)^{-\delta}\\ &+\bar{A}(R)\left[\frac{2\delta-1}{\delta-1}-\delta R\times I_{i=2}\right]\left(c+\xi_i+\frac{t}{S_i}\right)^{1-\delta}\\ &-(K+\Theta_i).\end{aligned} \tag{11}$$

We consider the following optimization problem:

$$\begin{aligned}Q^* &= \max_{(t,FC,R)\geq 0} Q(t,FC,R)\\ &\equiv \max_{(t,FC,R)\geq 0} Q_i(t,FC,R)\,|_{i=i^*(t,FC,R)},\end{aligned} \tag{12}$$

where $i^*(t,FC,R)$ is the firm's optimal technology choice in response to (t,FC,R) as per section 3.1. Note that although Equation (11) does not explicitly depend on the value of FC (subsidy cancels[6] out, entering with a positive sign in the Firm's Profit term and a negative sign in the Consumer Surplus term), the optimal welfare Q^* depends on FC

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indirectly via its effect on the optimal technology choice $i^*(t, FC, R)$. Finally note that while the expressions above were derived for $k = 2$ technologies, the extension to the $k > 2$ case is straightforward.

In the remainder of the article we will derive and analyze several different forms of environmental policies for the regulator. We start by deriving the optimal welfare under *centralized* control, which will serve as a convenient benchmark for the other cases.

4.1. Centralized Solution

By the centralized solution (sometimes referred to as the "first best" in economics or "coordinated" solution in operations/supply chain literature) we mean the optimal welfare in a case where the regulator has direct control over not only the environmental policy but also over the firm's response to it, that is, can dictate the price, production quantity, and production technology. Clearly, this type of solution, while not very realistic, will achieve the highest possible level of social welfare, providing a convenient upper bound on the value of social welfare Q^* in the Stackelberg game. It is also clear that in this case there is no need for taxes, subsidies, or rebates since the regulator can directly control the decisions these instruments are designed to influence. Thus the welfare in the centralized solution, which we denote by superscript "(C)" is obtained by computing $Q^{*(C)} = \max_{p \geq 0, i=1,\ldots,k} Q_i^{(C)}(p)$, where

$$\begin{aligned} Q_i^{(C)}(p) &= Firm'sProfit + \text{Consumer Surplus} \\ &\quad - \text{Environmental Impact} \\ &= \left(p - \left(c + \xi_i + \frac{\varepsilon}{S_i}\right)\right) D(p) - \Theta_i - K \\ &\quad + \int_p^\infty D(w)dw. \end{aligned} \tag{13}$$

It is not difficult to verify that Equation (13) is unimodular in p and is optimized at

$$p_i^{*(C)} = c + \xi_i + \frac{\varepsilon}{S_i}, \tag{14}$$

that is, the optimal price = marginal societal cost, as one would expect for the central planner case. Substituting this expression into Equation (13) we obtain

$$Q_i^{*(C)} = \frac{A(c + \xi_i + \varepsilon/S_i)^{1-\delta}}{\delta - 1} - \Theta_i - K, \tag{15}$$

with $Q^{*(C)} = \max_{i=1,\ldots,k}\{Q_i^{*(C)}\}$; this derivation is not limited to $k = 2$.

Note that if the environmental concern parameter $\varepsilon = 0$, the social welfare $Q^{*(C)}$ is maximized at technology 1: when the society does not care about the environmental damage, the dirtiest technology will be selected. As ε increases, the maximum social welfare $Q_i^{*(C)}$ declines for each technology i, but the rate of decline depends on the technological characteristics (Θ_i, ξ_i, S_i). In fact, ε here plays a similar role to the tax rate t in the firm's problem analyzed in section 3.1. This is evidenced by the following result for the case of two technologies (the proof follows directly from Equation (15) and is omitted).

Proposition 4. *For the centralized model with two technologies $k = 2$ and $S_1 < S_2$,*

(i) There exist $0 \leq \varepsilon_{12}^{crit} < \varepsilon_{21}^{crit}$ (with, possibly, $\varepsilon_{21}^{crit} = \infty$) such that the cleaner technology 2 is optimal iff $\varepsilon \in [\varepsilon_{12}^{crit}, \varepsilon_{21}^{crit}]$, and the dirtier technology 1 is optimal otherwise.

(ii) For the equal fixed cost case $\Theta_1 = \Theta_2$, $\varepsilon_{21}^{crit} = \infty$, that is, technology 2 is optimal whenever $\varepsilon \geq \varepsilon_{12}^{crit} = \frac{(\xi_2 - \xi_1)S_1S_2}{S_2 - S_1}$.

That is, the dirtier technology will be selected if the environmental concern parameter is either too low (below ε_{12}^{crit}) or too high (above ε_{21}^{crit}), with the second case being similar to the non-monotone technology choice effect of section 3.1: in both cases, the optimal prices are set so high that the demand falls below the level required to justify the capital investments in the cleaner technology. Moreover, this effect disappears when the fixed costs of the two technologies are the same.

5. Socially Optimal Environmental Policy

In this section, we analyze the socially optimal environmental policy and the corresponding socially optimal technology choice. The "brute force" analysis of such a policy is quite complex because it involves optimization of a non-convex and discontinuous function (as will be shown below) over three variables. Therefore we first consider two simplified *policy structures*: the tax-only policy, that is, $(t,0,0)$, and the tax-and-subsidy policy, that is, $(t,FC,0)$. With the insight obtained from the analysis of the simplified policies we then derive the combined jointly optimal socially optimal policy (t,FC,R).

5.1. Analysis of the Tax-Only Policy

In this subsection, we focus on the impact of the environmental tax t, assuming that the subsidy and rebate values are fixed (same as in the firm's problem w.l.o.g. we can assume $FC = R = 0$). Thus we suppress FC and R in all expression in this subsection and use superscript "(T)" to identify expressions applicable to the tax-only policies (to differentiate these from

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expressions for other policy structures discussed later in the article).

Observe that the function $Q_i^T(t) \equiv Q_i(t,0,0)$ defined by Equation (11) is quasi-concave and has a unique maximum at t_i^{opt} defined by

$$\begin{aligned} \tilde{t}_i^{opt} &= \frac{(\delta-1)\ \varepsilon - S_i(c+\xi_i)}{\delta}; \\ t_i^{opt} &= \min\{\max\{\tilde{t}_i^{opt},0\}, t_i^{lim}\} \end{aligned} \tag{16}$$

(the second part of the expression ensures the tax rate stays in the relevant range). Several interesting observations can be made. First, if the environmental concern parameter ε is small enough, then $t_i^{opt} = 0$ for all i, implying that the regulator will not tax the emissions at all, inducing the firm to choose the dirty technology 1 (by Assumption 2). It is straightforward to verify from Equation (16) that for taxation to have a possibility of motivating the choice of technology i, the regulator (or society) must weigh the environmental impact at a certain minimal level, as indicated in the following result.

Proposition 5. *The optimal tax $t^{*(T)} = 0$ and the dirtiest technology 1 will be chosen if*

$$\varepsilon \leq \min_{i=1,\ldots,k} S_i(c+\xi_i)/(\delta-1).$$

Second, observe that t_i^{opt} is increasing in ε. This is intuitive—the regulator that is more concerned with the environmental impact should naturally have a tendency to set higher taxation levels. Note, however, that the actual social welfare function $Q^{(T)}(t)$, specified in Equation (12), may not be maximized at any of the t_i^{opt} values: these values may lie outside the intervals over which the corresponding technologies are selected (i.e., it may happen that $i^*(t_i^{opt}) \neq i$ for all i). To find the optimal tax level, we must therefore consider the firm's technology choice, which from the earlier discussion changes from i to j at t_{ij}^{crit}.

An important observation is that unlike the firm's profit, which is identical regardless of whether t_{ij}^{crit} is approached from the left of right (by the definition of the critical taxation level $Pr_1(t_{ij}^{crit}) = Pr_2(t_{ij}^{crit})$ for $i \neq j \in \{1,2\}$), the welfare function may be discontinuous at t_{ij}^{crit}. Indeed, if the critical level is approached from below, the production quantity will approach $x_i^*(t_{ij}^{crit})$, while approaching the same critical level from above will result in the production level of $x_j^*(t_{ij}^{crit})$–which may be very different. This leads to different emission levels and other differences in the components of the social welfare function. Thus $Q_1^{(T)}(t_{ij}^{crit}) \neq Q_2^{(T)}(t_{ij}^{crit})$, leading to the discontinuity of $Q^{(T)}(t)$ at t_{ij}^{crit}.

To avoid this indeterminacy, we therefore replace each critical tax level t_{ij}^{crit} with two values: t_{ij}^{crit-}, the limit as the critical value is approached from the left (below), and t_{ij}^{crit+}, the limit as the same value is approached from the right (above). We treat t_{ij}^{crit-} and t_{ij}^{crit+} as two different rates. In practice, we can assume that there exists some small value $\mu > 0$ such that $t_{ij}^{crit-} = t_{ij}^{crit} - \mu$ and $t_{ij}^{crit+} = t_{ij}^{crit} + \mu$.

With these definitions, the choice region for technology 2 becomes $t \in [t_{12}^{crit+}, t_{21}^{crit-}]$ and for technology 1 the corresponding region is $t \in [0, t_{12}^{crit-}] \cup [t_{21}^{crit+}, t_1^{lim}]$. This leads to the following characterization of the optimal taxation levels.

Proposition 6. *Define two tax levels $t_1^{*(T)}$ and $t_2^{*(T)}$ as follows:*

(i) If $t_1^{optT} \in [0, t_{12}^{crit-}] \cup [t_{21}^{crit+}, t_1^{lim}]$ then $t_1^{*(T)} = t_1^{opt}$. Otherwise, $t_1^{*(T)} = \arg\max\{Q_1^{(T)}(0), Q_1^{(T)}(t_{12}^{crit-}), Q_1^{(T)}(t_{21}^{crit+}), Q_1^{(T)}(t_1^{lim})\}$.

(ii) If $t_2^{opt} \in [t_{12}^{crit+}, t_{21}^{crit-}]$, then $t_2^{*(T)} = t_2^{opt}$. Otherwise, $t_2^{*(T)} = \arg\max\{Q_2^{(T)}(t_{12}^{crit+}), Q_2^{(T)}(t_{21}^{crit-})\}$.

Under the tax-only policy structure the optimal value of the social welfare is $Q^{*(T)} = \max\{Q_1^{(T)}(t_1^{*(T)}), Q_2^{(T)}(t_2^{*(T)})\}$ and the optimal tax is $t^{*(T)} = \arg\max\{Q_1^{(T)}(t_1^{*(T)}), Q_2^{(T)}(t_2^{*(T)})\}$.

This proposition presents an important technical result: although the welfare function is discontinuous and not quasi-concave, the optimal tax level in the case with $k = 2$ technologies can be found by examining at most eight points. A similar procedure can be developed for the $k > 2$ technology as well: the optimal tax rate can be found by examining a discrete set of points of the size $O(k)$; see the Appendix.

We also make three observations based on the preceding result. First, if technology choice $i = i^*(t^{*(T)})$ is optimal under the tax-only policy, the optimal tax rate is achieved at either t_i^{opt} or at one of the breakpoints, which include the critical tax levels, as well as $t = 0$ and t_1^{lim}. These two situations are quite different: unlike t_i^{opt}, which is increasing in the environmental concern parameter ε, the breakpoints are independent of this parameter. Thus (as will be seen in the example below), once a certain technology is optimal, it typically remains so for a wide ranges of ε, ensuring that the model is robust with respect to this parameter, which, arguably, is difficult to estimate exactly in practice.

Second, some of the possible values for $t^{*(T)}$, while being optimal in the strict sense defined above, may be difficult for the regulator to justify politically. For example, if $t^{*(T)} = t_{12}^{crit-}$, then it is optimal to set the environmental tax at the maximum level that will

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ensure that the switch to the cleaner technology 2 does *not* occur. This may be hard to justify as an "environmental" tax.

Third, the optimal value of the social welfare and the optimal tax rate are both affected by the environmental concern parameter ε. From Proposition 5, when ε is low enough, technology 1 is optimal. As ε increases, the relative attraction of technology 2 grows. However, whether it is induced at the socially optimal tax rate or not depends on the characteristics of the technology. It may also happen that even if technology 2 is induced for some range of values of ε, once the environmental concern parameter exceeds these values, technology 1 may become socially optimal again (this happens because the optimal tax rate is pushed into the region where the non-monotone technology choice occurs). This interplay between the values of ε and the model outcomes is illustrated in the following example.

EXAMPLE 2. Continuing from Example 1, Figure 4 presents the plots of the total social welfare as a function of tax level for the cases with $\varepsilon = 0.3$, 0.6, 1.2, 3, 4.5, 6. Case (a) illustrates the observation that when ε is too small, the optimal tax will be zero (the bold line, denoting $Q^{(T)}$ is highest at $t = 0$). In (b) with $\varepsilon = 0.6$ the regulator is not particularly concerned with the environmental issues and therefore the optimal tax level $t^{*(T)} = t_1^{opt} = 0.07$ is low, which motivates the firm to choose the dirty technology 1. In (c) through (e) the regulator is moderately concerned with the environmental issues ($\varepsilon = 1.2,3,4.5$), and therefore the optimal tax rate is sufficiently high to motivate the firm to choose the

Figure 4 Social Welfare Under the Tax-Only Policy as a Function of Tax Rate for ε = 0.3, 0.6, 1.2, 3, 4.5, 6.

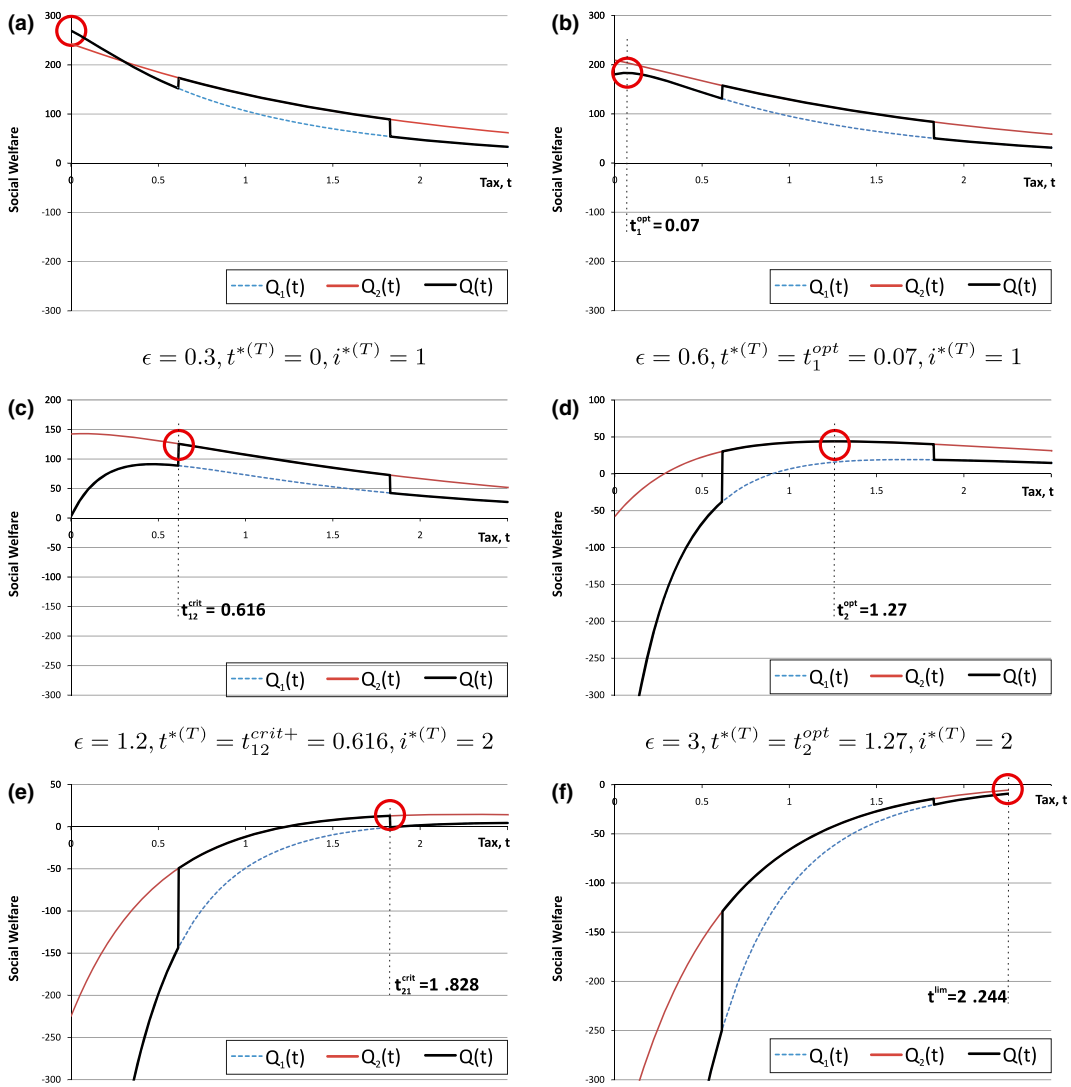

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Table 1 The Optimal Tax Level and Technology Choice under the Tax-Only Policy for Different Values of ε

Interval of ε values	0–0.5	0.5–0.8	0.8–2	2–3.9	3.9–5.2	>5.2
Socially optimal tax level	0	$t^{*(T)} = t_1^{opt}$	$t^{*(T)} = t_{12}^{crit+}$	$t^{opt(T)} = t_2^{opt}$	$t^{*(T)} = t_{21}^{crit-}$	$t^{*(T)} = t_1^{lim}$
Technology choice	Dirty	Dirty	Clean/boundary	Clean	Clean/boundary	Dirty

clean technology 2. In (f) the regulator is extremely concerned with the environmental issues ($\varepsilon = 6$), which leads to the very high optimal tax level, inducing the firm to select the dirty technology.

The relationship between the values of ε, socially optimal tax level, and technology choice for this example is summarized in Table 1, as well as in Figure 5. Observe that in this case for $0.8 \leq \varepsilon \leq 5.2$, the clean technology is induced and is socially optimal, with technology 1 being socially optimal when $\varepsilon < 0.8$ or $\varepsilon > 5.2$ (at $\varepsilon = 0.8$ or $\varepsilon = 5.2$ the optimal tax rate is not unique). This illustrates that high levels of society's environmental concerns do not necessarily lead to a cleaner technology choice. In this case, when ε is very high the socially optimal tax rate is also very high, which increases the equilibrium prices, reducing firms' revenues and making it impossible to recoup the higher installation cost of technology 2.

Note that the ranges of ε values for which the technology choice is invariant are quite wide, indicating that the model is robust with respect to ε—which is important since estimating the "correct" value of ε may be difficult in practice.

Since the parameter ε can be interpreted as the marginal value of the damage from the environmental pollution, it may be tempting to assume that the optimal environmental tax level should be at least as high, that is, that $t^{*(T)} \geq \varepsilon$ should hold. However, this logic is not correct: Figure 5 shows that $t^{*(T)} < \varepsilon$ for all values of ε in the example above. This is because the environmental tax level influences not just emissions but also all other components of the social welfare function. In fact, it is not hard to show that the following result.

Proposition 7. *Let ε_{ij} be the smallest ε such that $t^{*(T)} = t_{ij}^{crit+}$. If $\varepsilon_{ij} > t_{ij}^{crit+}$, then for any $\varepsilon > 0$, the optimal tax level under the tax-only policy $t^{*(T)} \leq \varepsilon$.*

We note that setting $t = \varepsilon$ may be optimal under a more general environmental policy, as will be discussed in section 5.3. However, we first discuss how the welfare under the tax-only policy compares with the centralized upper bound.

5.1.1 Comparison between the Tax-Only and Centralized Solutions. It is easy to verify that under the the tax-only policy structure the regulator finds it optimal to set $t^{*(T)} = t_i^{opt}$ as given by Equation (16) for some technology i and $t_i^{opt} \in (0, t_i^{lim})$, then substituting the tax rate from Equation (16) into the expression for the optimal price in Equation (4), we obtain the expression identical to the optimal price in the centralized solution given by Equation (14). This immediately leads to the following result:

Proposition 8. *Assume $t_i^{opt} \in (0, t_i^{lim})$ for all i. Then for the centralized solution $Q^{*(C)} = \max_{i=1,\ldots,k} Q_i^{(T)}(t_i^{opt})$. Moreover, suppose the maximum is achieved for some technology j. Then $Q^{*(C)} = Q^{*(T)}$ (i.e. the centralized and tax-only solutions are the same) iff $t^{*(T)} = t_j^{opt}$.*

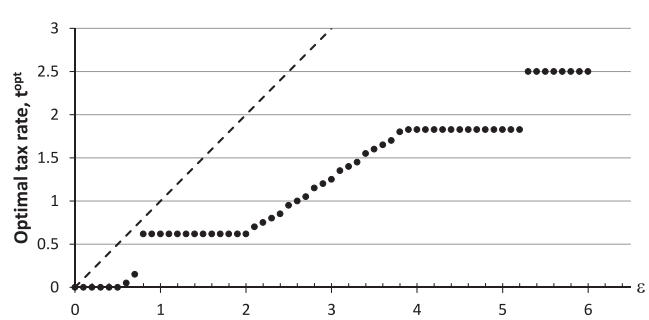
This proposition presents an important insight: were the regulator able to "dictate" the firm's technology choice, the tax-only environmental policy would achieve the highest possible social welfare. As we will see below, the use of fixed cost subsidies and consumer rebates enables the regulator to essentially dictate the technology choice in most cases.

We also note that it is not true, in general, that for the same parameter values a cleaner technology will be induced in the centralized vs. the tax-only case. For example, in the two-technology case if $Q_1^{(T)}(t_1^{opt}) > Q_2^{(T)}(t_2^{opt})$, then the dirtier technology will be socially optimal in the centralized case, while the cleaner one may be optimal under the tax-only policy. Of course, it is easy to construct examples where the reverse occurs as well.

5.2. Analysis of the Tax-and-Subsidy Policy

In this section, we consider a class of policies combining taxes and fixed cost subsidies, that is, policies of the form $(t,FC,0)$. We refer to these as *tax-and-subsidy* policies and denote the relevant expressions with the

Figure 5 Optimal Tax Level (Solid Dots) under the Tax-Only Policy for Different Values of ε (the $t = \varepsilon$ Line is Plotted with Dashes)



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superscript "(*FC*)". Recall that by offering the fixed cost subsidy $FC > 0$ when the firm selects technology 2, the regulator reduces the installation cost of this technology to $\Theta_2 - FC$, while the cost of technology 1 remains unchanged. As noted in section 4, the subsidy has no effect on the technology-specific welfare functions $Q_i^{(FC)}(t, FC) \equiv Q_i(t, FC, 0) = Q_i^{(T)}(t)$, thus also not affecting their maximizers t_i^{opt} in Equation (16).

However, the value of *FC* does have an impact on the technology choice $i^*(t, FC) \equiv i^*(t, FC, 0)$. Let $t_{ij}^{crit}(FC)$ be the critical taxation level when subsidy FC is offered. From section 3.1 $i^*(t, FC) = 2$ if $t \in [t_{12}^{crit+}(FC), t_{21}^{crit-}(FC)]$, and by Corollary 1 the critical taxation levels change and the interval above expands with *FC*; see Figure 3 for an illustration. In fact, by offering a large enough subsidy (possibly larger than Θ_2) it is always possible to ensure that $t_2^{opt} \in [t_{12}^{crit+}(FC), t_{21}^{crit-}(FC)]$. Thus $Q_2^{*(FC)} = Q_2^{(T)}(t_2^{opt})$. Note that the value of *FC* for which the relationship above holds is not unique—in fact, if the equality holds for some $FC' \geq 0$, it also holds for any $FC \geq FC'$. Thus we define FC^* as the minimum value of subsidy at which the social welfare is maximized. We have the following result.

Proposition 9. *The optimal environmental policy under the tax-and-subsidy structure is the following:*

(i) If $Q^{*(T)} = Q_2^{(T)}(t_2^{opt})$, that is, the welfare under the tax-only policy structure is optimized at t_2^{opt}, then $FC^* = 0$, $t^{*(FC)} = t^{*(T)} = t_2^{opt}$, $Q^{*(FC)} = Q_2^{(T)}(t_2^{opt})$, and $i^{*(FC)} = 2$.

(ii) If $Q^{*(T)} < Q_2^{(T)}(t_2^{opt})$, then there exists $FC^* = \arg\min_{FC>0}\{t_2^{opt} \in [t_{12}^{crit}(FC), t_{21}^{crit}(FC)]\}$, that is, FC^* is the minimal subsidy such that technology 2 is induced at tax rate t_2^{opt}. In this case, $t^{*(FC)} = t_2^{opt}$, $Q^{*(FC)} = Q_2^{(T)}(t_2^{opt})$, and $i^{*(FC)} = 2$.

(iii) If $Q^{*(T)} > Q_2^{(T)}(t_2^{opt})$, that is, the welfare under the tax-only policy structure is optimized with technology 1, then $FC^* = 0$, $t^{*(FC)} = t^{*(T)}$, $Q^{*(FC)} = Q^{*(T)}$, and $i^{*(FC)} = 1$.

This proposition underscores the role of fixed cost subsidies in our model: they essentially allow the regulator to achieve $Q_2^{(T)}(t_2^{opt})$—the maximum possible social welfare in case technology is set to 2 under the tax-only policy. If this is achieved automatically (case (i) above) or would be counter-productive (case (iii) above), then no subsidy should be used. Otherwise, offering a fixed cost subsidy improves the social welfare above the tax-only policy. Moreover, if case (ii) holds and $Q_2^{(T)}(t_2^{opt}) > Q_1^{(T)}(t_1^{opt})$, then, by Proposition 8, $Q^{*(FC)} = Q^{*(C)}$, that is, the optimal tax-and-subsidy policy leads to the centralized solution. Continuing with case (ii), we also note that if $t_2^{opt} > t^{*(T)}$, then from Proposition 3 and Corollary 1 we know that $FC^* < \Theta_2$; otherwise FC^* could be larger than the fixed cost of the clean technology. Which of these three cases occurs depends on the values of various model parameters; we investigate this issue numerically in section 5.4.

5.3. Consumer Rebates and Joint Environmental Policy

In this section, we add consumer rebates to our analysis, resulting in the general environmental policy (t, *FC*,*R*) consisting of taxes, fixed cost subsidies, and consumer rebates. We refer to these policies as "joint", designating the relevant expressions with a superscript ("*J*").

Recall that $R \in [0,1)$ represents the percentage rebate that the regulator offers to consumers who purchase the product if the firm chooses the clean technology 2. Providing a rebate has two effects. First, rebate expands the market size for technology 2 from A to $A(R) = A(1 - R)^{-\delta}$ so that the interval where this technology is induced expands, that is, the critical taxation level $t_{12}^{crit}(R)$ is non-increasing in R and $t_{21}^{crit}(R)$ is non-decreasing in R, acting analogously to fixed cost subsidies. Similarly, because the market size increases while the costs remain the same, providing rebates also widens the feasible range for the tax rate; t_2^{lim} is increasing in A and hence in R.

Second, and perhaps more importantly, rebate allows the regulator to essentially control the consumer price when the firm uses technology 2. From Equations (4) and (14) if the regulator sets $R = 1/\delta$ and $t = \varepsilon$, then the price seen by consumers equals

$$p_2^*(t) \times (1 - R) = p_2^*(\varepsilon)\frac{\delta - 1}{\delta} = c + \xi_2 + \frac{\varepsilon}{S_2} = p_2^{*(C)}.$$

That is, if the firm chooses technology 2, then the price seen by consumers under the policy (ε,*FC*,$1/\delta$) for any $FC \geq 0$ is the same as in the centralized solution, which means the equilibrium production quantity is also the same. This, in turn, implies that the total emissions, firm's profit, and consumer surplus must be the same as in the centralized solution, implying $Q_2(\varepsilon, FC, 1/\delta) = Q_2^{*(C)}$, that is, the upper bound on welfare (with technology 2) is achieved by offering a combination of tax and rebate and any level of fixed cost subsidy, including $FC = 0$. While technology 2 may not be selected with policy (ε, 0, $1/\delta$), as we saw in the previous section we can always induce it by providing a large enough fixed cost subsidy FC^R, where FC^R is the minimum value of fixed cost subsidy such that $i^*(\varepsilon, FC^R, 1/\delta) = 2$. This immediately leads to the following result.

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PROPOSITION 10. *The optimal environmental policy that combines tax, fixed cost subsidy, and consumer rebate is the following:*

(i) If $Q_2^{*(C)} \geq Q_1^{*(T)}$, that is, the optimal welfare obtained when technology 2 is selected in the centralized case is higher than the optimal welfare obtained with technology 1 under the tax-only policy, then the optimal environmental policy $(t^{*(J)}, FC^{*(J)}, R^{*(J)}) = (\varepsilon, FC^R, 1/\delta)$, and the optimal social welfare is $Q^* = Q_2(\varepsilon, FC^R, 1/\delta) = Q_2^{*(C)}$. Furthermore, if $= 4Q_2^{*(C)} \geq Q_1^{*(C)}$; that is, technology 2 is optimal in the centralized solution, then from Proposition 8 the centralized and the jointly optimal solutions are the same.

(ii) If $Q_2^{*(C)} < Q_1^{*(T)}$ then the optimal joint environmental policy coincides with the tax-only policy, that is, $(t^{*(J)}, FC^{*(J)}, R^{*(J)}) = (t^{*(T)}, 0, 0)$, and the optimal welfare $Q^* = Q_1(t^{*(T)}, 0, 0)$. In this case, from Proposition 8, if $t^{*(T)} = t_1^{opt}$, then the centralized and the jointly optimal solutions are also the same.

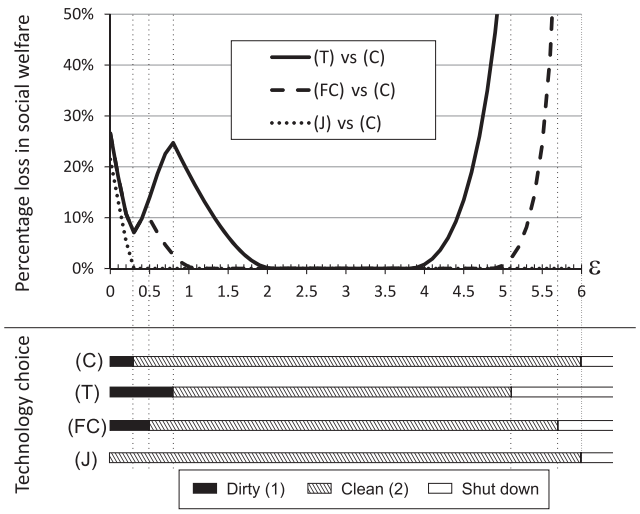
To summarize, using a combination of the environmental tax, consumer rebate, and fixed cost subsidy, the regulator can always induce technology 2 and achieve the highest possible welfare value of social welfare achievable with this technology (the same that would be achieved for this technology in the centralized case). The range of parameter values for which technology 2 is induced when social welfare is optimized using the joint policy is significantly wider than in cases where only the taxation or the tax and fixed cost subsidies are used. This applies to many cases where the value placed by society on pollution (i.e., ε) is either too low or too high to induce technology 2 using taxation alone. We also note that technology 2 may be optimal under the joint optimal policy even when it is not optimal in the centralized system.

5.4. Comparisons between Policies: Numerical Experiments

In this section, we compare the welfare and the technology choice obtained with the three policy structures: tax-only (*T*), tax-and-subsidy (*FC*), combined/joint (*J*), in comparison with the centralized upper-bound (*C*). To do so we performed two numerical experiments.

In the first experiment we used the parameter settings in Example 2 above ($A = 1{,}000{,}000, \delta = 3, c = 1, K = 12{,}000, S_1 = 1, \xi_1 = \Theta_1 = 0, S_2 = 2, \xi_2 = 0.1, \Theta_2 = 18{,}000$) and varied the environmental concern parameter, ε. For each ε we calculated the optimal social welfare for each policy class and measured the relative difference to the optimal value of the (*C*) solution. We have also recorded the optimal technology choice in each case. The results are reported in Figure 6. The following observations can be made for different ranges of ε:

Figure 6 Percentage Improvement in Optimal Value of Social Welfare for the Centralized (*C*), Tax-Only (*T*), Tax-and-Subsidy (*FC*), and Joint (*J*) Solutions over the Decentralized Solution



(1) For very low values of ε, i.e., $\varepsilon \in [0,0.3)$, the (*C*) solution selects technology 1. The (*T*) policy also selects technology 1 and has a gap of up to 26.5% vs. the (*C*) solution. This gap cannot be closed by either (*FC*) or (*J*) policies, which select technologies 1 and 2, respectively. In this range the society does not place enough value on the pollution prevention, and so the higher-cost technology 2 is not competitive.

(2) For $\varepsilon \in [.3,1]$, the (*C*) solution switches to technology 2. The gap for the (*T*) policy remains substantial, up to 25%. For most of this range (up to $\varepsilon < 0.8$) the tax-only solution induces technology 1; even when technology 2 is induced, the optimal tax rate is at the critical point t_{12}^{crit+}, which is below t_2^{opt}. The (*FC*) policy closes some but not all of the gap, and for most of this range ($\varepsilon > 0.5$) induces technology 2, reducing it to < 10% in all cases. The optimal value of FC in these cases far exceeds the installation cost Θ_2 of technology 2. The (*J*) policy eliminates the gap completely using technology 2, offering rebates ranging from 16% to 64% and using no fixed cost subsidies.

(3) For $\varepsilon \in [1.1,2]$, all solutions induce technology 2. The gap vs. (*C*) is < 10% for the (*T*) policy and is eliminated entirely by the (*FC*) and (*J*) policies.

(4) For $\varepsilon \in [2.1,3.8]$, all solutions coincide, inducing technology 2. Thus the tax-only strategy (*T*) is sufficient to achieve system

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coordination; there is no need for either fixed-cost subsidies or consumer rebates.

(5) For $\varepsilon \in [3.9, 4.8]$, all solutions continue to induce technology 2. A gap, growing rapidly with ε, opens between the (*T*) and (*C*) solutions: the high level of society's concern for pollution is pushing the tax to high levels, where t_2^{opt} is beyond the range where technology 2 is induced. Both (*FC*) and (*J*) solutions close the gap to (*C*), indicating that tax-and-subsidy policy is sufficient to coordinate the system.

(6) For $\varepsilon \in [4.9, 5.1]$, all solutions continue to induce technology 2. The gap for (*T*) vs. (*C*) continues to grow as the high taxes induce operations further away from the optimal level by the firm. At $\varepsilon = 5.1$ the optimal social welfare in the tax-only system becomes negative, indicating that it is better for the regulator to set the tax rate above t^{lim}, causing a shutdown by the firm and bringing social welfare up to 0. In this range, (*FC*) closes some, but not all of the gap to (*C*). The gap to (*C*) is closed by the (*J*) policy.

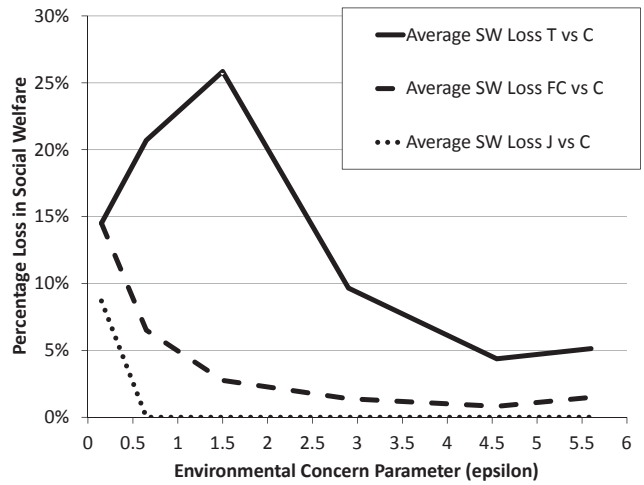
(7) For $\varepsilon \in [5.2, 5.9]$, the tax-only system is no longer functioning (the "optimal" tax rate induces technology 1, but leads to the negative value of social welfare). The (*FC*) strategy also loses traction, with the optimal value of social welfare becoming negative at $\varepsilon = 5.7$, causing a shutdown as well. The centralized (*C*) solution continues to induce technology 2, and the combined strategy (*J*) produces the same solutions.

(8) For $\varepsilon > 6$ the optimal social welfare under for the centralized system (*C*) becomes negative, indicating that system shutdown is optimal in this range.

To summarize, except for the case of very low values of ε, it was possible to achieve system coordination by the general environmental policy (*J*) in all cases (it should be noted that the (*J*) policy never involved positive fixed cost subsidies for this example—consumer rebates and taxes were sufficient). On the other hand, if the regulator is restricted to using only the environmental taxes, then the gap vs. (*C*) solution can be substantial, except for a range of moderately high ε where the system self-coordinates. The range of values where tax-and-subsidy policies achieved system was wider compared with the tax-only policies, but narrower compared with the joint policies.

In the second experiment, we verified the above insight in a wider set of numerical simulations. To do so we sampled the model parameters from the following ranges: $\delta \in (1, 5], S_2 \in [1, 20], \xi_2 \in [0, 1], \Theta_2 \in [10{,}000, 20{,}000]$, $\varepsilon \in [0, 6]$ while keeping the base case parameters the same as in the preceding example.

Figure 7 Results of the Simulation: Average Welfare Loss in the Stackelberg Solutions as a Function of ε



Using the uniform distributions (independently for each parameter) we sampled 1000 instances from this hypercube of values and observed the performance of the three policies (*T*, *FC*, *J*) vs. the (*C*) solution.

In 293 of 1000 parameter combinations in solutions (*C*, *FC*, *J*) it was optimal to shut down production, resulting in the optimal welfare of zero, and in 29 additional combinations shutdown was optimal under the tax-only policy. Hence, to have a non-trivial basis for comparison we removed those cases, resulting in 678 instances that we compared.

Figure 7 presents the average percentage loss in social welfare of the (*T*, *FC*, *J*) policies as compared with the centralized solution (*C*) as a function of the environmental concern parameter, ε. The pattern described for the first set of experiments above is clearly confirmed: unless ε is very small, the joint policy (*J*) reaches the centralized upper-bound on welfare, coordinating the system (clearly the clean technology is induced in these cases). Tax-only policy (*T*) performs worst when ε is low, but not very low: in this case the average welfare loss could reach some 25%. As suggested by the analytical results above, this happens in the range where society's environmental concerns are either not high enough to induce the clean technology choice with the tax-only policy or, even if the clean technology choice is induced, the resulting optimal tax rate is at a critical point that is far away from the value that would maximize the social welfare under technology 2. The tax-and-subsidy policy (*FC*) performs much better than the tax-only policy, leading to welfare losses of < 5% in most cases. This indicates that the tax-and-subsidy policy may be a good policy alternative when the implementation of consumer rebates is not feasible for political reasons or expensive because of the transactional costs of processing consumer rebates.

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It is also interesting to note that there were 40 instances where the centralized (*C*) solution selected the dirty technology 1 due to very small ε (in those instances the average value of $\varepsilon = 0.13$ and the largest is 0.31). While the (*T*) and (*FC*) policies also selected technology 1 in all those cases, the joint policy (*J*) selected the clean technology in 24 of 40 cases. That is, the Stackelberg game solution could actually lead to a cleaner technology choice than the centralized solution.

6. Conclusions and Future Research

In this study, we examined the role that environmental taxation can play in reducing environmental pollution and inducing the choice of greener technology by a profit-maximizing firm. We showed that, in general, the firm's response to taxation may be non-monotone: higher levels of taxes may induce dirtier rather than cleaner technology. We also modeled the regulator's problem as the optimization of social welfare consisting of consumer surplus (which includes net tax revenues that by assumption are "returned" to consumers either in the form of government services or direct refunds), firm's profit, and the total pollution, the latter weighed by parameter ε, which represents societal environmental concerns. We showed that the effect of very high levels of ε, may be similar to high levels of taxes: the socially optimal choice may be a dirtier technology with reduced production levels.

We compared the original decentralized system (modeled as a Stackelberg game) to the coordinated system where the regulator has control over prices, technology choice, and production levels. When the decentralized system achieves the same level of social welfare as the centralized system, the regulator has achieved the theoretical upper bound by coordinating the system. We showed that while the environmental taxes alone may not be sufficient to coordinate the system, in many cases such coordination can be achieved by the use of a strategy that combines environmental taxes with two other policy tools: fixed cost subsidies and consumer rebates. The latter are particularly effective—they are often sufficient to induce coordination without the use of subsidies.

In terms of future work, the immediate extension that we address in the follow-up study is the oligopolistic case with multiple heterogenous firms competing in a Cournot fashion. Due to the technical challenges, this work mostly focuses on the firms' reactions to a given tax rate; the analysis of the regulator's problem remains open. Other extensions could explicitly incorporate uncertainty and asymmetry of information, both in demand and in technologies' cost or environmental performance, as well as to explicitly consider how motivating the adoption of the clean technologies propagates down the "innovation chain," motivating the firms that develop green technologies. It is also important to explore the practical ways of estimating the environmental concerns parameter ε, and perform the corresponding empirical studies.

Appendix A. Proofs

Proof of Proposition 1. From Equations (6) and (7) the equation $\frac{\partial \Delta(t)}{\partial t} = 0$ has a unique root given by the first expression in the minimum in Equation (8). Further, at $t = 0$ the derivative

$$\frac{\partial \Delta(t)}{\partial t} = \bar{A}(\delta - 1)\left(\frac{1}{S_1(c+\xi_1)} - \frac{1}{S_2(c+\xi_2)}\right) > 0$$

because $S_2 > S_1$ and $\xi_2 > \xi_1$. Hence $\Delta(t)$ is increasing for $t \in [0, t^{max})$ and is decreasing for $t \in (t^{max}, \infty)$, that is, is quasi-concave. □

Proof of Proposition 2. First observe that for A sufficiently large, $t_2^{lim} > t^{max}$. This follows because by definition t_2^{lim} is increasing in A, while from Equation (8) t^{max} is independent of A.

Let $\Theta = \Theta_2 - \Theta_1$ be the difference in fixed costs of technologies 1 and 2. From Equation (7) it is easy to see that $\Delta(t) = f(t) - \Theta$, where $f(t)$ is independent of Θ_1, Θ_2. Moreover, under assumption $\xi_1 < \xi_2$, $f(t)$ is a quasi-concave function increasing for $t \in (0, t^{max}]$. Thus, by selecting new values Θ_1', Θ_2' of fixed cost parameters so that $\Theta' = \Theta_2' - \Theta_1' = f(t^{max})$, we ensure that $\Delta(t^{max}) = 0$ and $\Delta(t) < 0$ for all $t > 0$, $t \neq t^{max}$.

Now, choose $\varepsilon > 0$ and suppose the fixed cost parameters are adjusted so that $\Theta'(\varepsilon) = \Theta' - \varepsilon$. Then $\Delta(t^{max}) = \varepsilon$, and $\Delta(t)$ has exactly two roots on $(0, \infty)$ provided ε is sufficiently small. Let $t_2^{crit}(\varepsilon)$ be the larger root. Note that $t_2^{crit}(\varepsilon) > t^{max}$ for $\varepsilon > 0$ and is a continuous function of ε. Thus, if the market capacity assumption $t^{max} < t_2^{lim}$ holds, there must exist $\varepsilon > 0$ such that $t^{max} < t_2^{crit}(\varepsilon) < t_2^{lim}$, implying that Region III, which equals to $t \in (t_2^{crit}(\varepsilon), t_2^{lim})$, is non-empty. To summarize, we have shown that under the assumptions of the proposition there always exist values of fixed cost parameters Θ_1, Θ_2 for which Region III is non-empty. □

Proof of Proposition 3. Note: This proof relies on the results for three-plus technologies, discussed in Appendix C.

Consider technologies k and $i \neq k$. Recall that, by assumption, $S_k > S_i$. Note that for arbitrary $y_k \leq y_i$, we have

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$$D(p)(p - y_i) \le D(p)(p - y_k)$$

for any $p \ge 0$. Taking maximum over $p \ge 0$ of both sides above, we see that $Pr_k(t) \ge Pr_i(t)$. Thus, $\Delta_{ki}(t) \ge 0$ if and only if

$$\xi_k + t/S_k \le \xi_i + t/S_i,$$

which is equivalent to $t \ge t_{ik}$ where the latter quantity is defined in the statement of the proposition. This implies that $t_{ik}^{crit} = \min\{t_k^{lim}, t_{ik}\}$ and $t_{ki}^{crit} = t_k^{lim}$, which, together with Proposition 13, imply part (i) of the proposition. Part (ii) now follows immediately by Proposition 13. □

Proofs of Propositions 4, 5, and 6. Follow immediately from the discussions preceding these propositions and hence are omitted. □

Proof of Proposition 7. Suppose that technology 1 is optimal. Then from Proposition 6: If $t^{*(T)} = t_1^{opt}$, then from Equation (16)$t^{*(T)} < \varepsilon$. If $t^{*(T)} = 0$, then the result obviously holds. If $t^{*(T)} = t_{12}^{crit-}$ then because Q_1^T is quasi-concave $t_{12}^{crit-} < t_1^{opt} < \varepsilon$. If $t^{*(T)} = t_{21}^{crit+}$, then the result holds by the condition of the proposition. Finally, if $t^{*(T)} = t_1^{lim}$, then the optimal welfare equals zero, which from Equation (11) can only hold when $t^{*(T)} < \varepsilon$.

Similarly, suppose that technology 2 is optimal. Then from Proposition 6: If $t^{*(T)} = t_2^{opt}$, then from Equation (16) $t^{*(T)} < \varepsilon$. If $t^{*(T)} = t_{12}^{crit+}$, then the result holds by the condition of the proposition. Finally, if $t^{*(T)} = t_{21}^{crit-}$ then because Q_2^T is quasi-concave $t_{21}^{crit-} < t_2^{opt} < \varepsilon$. □

Proofs of Propositions 8, 9, and 10. Follow immediately from the discussions preceding these propositions and hence are omitted. □

Appendix B. The Linear Demand Function Case

In this appendix we demonstrate that the non-monotonicity in the firm's response to the environmental tax also holds for the case when the firm's demand function is linear (as opposed to the exponential demand we considered earlier).

Let $D(p) = a - bp$. Then given technology choice, i, and tax level, t, the firm's profit function is

$$\begin{aligned} Pr_i(p,t) = &- bp^2 + (a + b(c + \xi_i + t/S_i)) \\ &p - (c + \xi_i + t/S_i)a - (K + \Theta_i), \end{aligned} \tag{A1}$$

which is maximized at

$$p_i^*(t) = \frac{a + b(c + \xi_i + t/S_i)}{2b}, \tag{A2}$$

with the associated profit level, $Pr_i(t)$, obtained by substituting Equation (A2) into Equation (A1) as follows:

$$\begin{aligned} Pr_i(t) &\equiv Pr_i(p_i^*(t), t) \\ &= \frac{(a + b(c + \xi_i + t/S_i))^2}{4b} - (c + \xi_i + t/S_i)a \\ &\quad - (K + \Theta_i). \end{aligned} \tag{A3}$$

It is easy to show that $Pr_i(t) \ge 0$ iff $0 < t \le t_i^{lim}$, where

$$t_i^{lim} = S_i\left(\frac{a}{b} - c - \xi_i\right). \tag{A4}$$

We assume that $t_i^{lim} > 0$ for all i, since otherwise technology i would never be chosen. Further for all technologies $i = 1,2,\ldots,k$ to be feasible, taxation level, t, must be bounded by t_i^{lim}, where

$$t^{lim} = \min_{i \in \{1,\ldots k\}} t_i^{lim}. \tag{A5}$$

As before, define

$$\Delta(t) = Pr_2(t) - Pr_1(t), \tag{A6}$$

meaning that if $\Delta(t) > 0$, then the firm would select a cleaner technology 2, and otherwise it would select a dirtier technology 1. By definition cleaner technology 2 is more expensive; in particular, we assume $\xi_2 \ge \xi_1$ and $\Theta_2 \ge \Theta_1$ with at least one inequality being strict.

Let $\alpha_i = b/4S_i^2$, $\beta_i \equiv t_i^{lim} = S_i(a/b - c - \xi_i)$, and $\gamma_i = K + \Theta_i$. Note that with our assumptions $\alpha_1 > \alpha_2$ and $\gamma_1 < \gamma_2$. With this notation, the equation $\Delta(t) = 0$ can be rewritten as

$$\alpha_2(t - \beta_2)^2 - \gamma_2 - \alpha_1(t - \beta_1)^2 + \gamma_1 = 0, \tag{A7}$$

and therefore if

$$\gamma_2 - \gamma_1 < \frac{\alpha_1\alpha_2(\beta_1 - \beta_2)^2}{\alpha_1 - \alpha_2}, \tag{A8}$$

then Equation (A7) has two roots (there is only one root if (A8) holds as an equality, and there are no roots if the opposite strict inequality holds). Simple algebra shows that these two roots are

$$t_{21,12}^{crit} = \frac{\alpha_1\beta_1 - \alpha_2\beta_2 \pm \sqrt{\alpha_1\alpha_2(\beta_1 - \beta_2)^2 - (\alpha_1 - \alpha_2)(\gamma_2 - \gamma_2)}}{\alpha_1 - \alpha_2}.$$

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As before, to conclude we show that there exists a set of parameters for which these two roots are feasible, that is, the roots themselves and the profits obtained by the firm at these roots are non-negative.

PROPOSITION 11. *If $\beta_2 > \beta_1$ and $\alpha_1\beta_1 > \alpha_2\beta_2$, then there exist γ_1 and γ_2 such that $0 < t_{12}^{crit} < t_{21}^{crit} < t_2^{lim}$.*

First, as we have just showed, roots $t_{12}^{crit} < t_{21}^{crit}$ exist if Equation (A8) holds. These roots are positive if $\Delta(0) < 0$ and $\Delta'(0) > 0$. The second inequality holds if $\alpha_1\beta_1 > \alpha_2\beta_2$ which is assumed by the condition of the proposition. The first inequality holds when $\alpha_2\beta_2^2 - \alpha_1\beta_1^2 < \gamma_2 - \gamma_1$, which does not contradict the existing conditions. Lastly, with the assumed conditions, it can be shown that $t_2^{lim} = \beta_2 - \sqrt{\gamma_2/\alpha_2}$, and thus choosing γ_2 such that $\gamma_2 < \alpha_2(t_2^{crit} - \beta_2)^2$ together with the above conditions ensures that both roots exist, are positive, and do not exceed t_2^{lim}, that is, the firm attains a positive profit in each of the three critical regions.

Note that only a single root and hence only two regions exist if $\Theta_1 = \Theta_2$; this follows from Proposition 3, which holds for a general demand function.

Appendix C. Technology Choice with $k \geq 3$ Technologies

The case with more than two technologies is similar, but somewhat more complex, than the two-technology case. We use the same notation as in the preceding section, adding indices designating technologies being compared. For example, $\Delta_{ji}(t) = Pr_j(t) - Pr_i(t)$ is the difference in profitability of technologies j and i at the taxation level t, and t_{ji}^{max} is the tax level that maximizes $\Delta_{ji}(t)$.

Our primary interest is in analyzing when the cleanest technology k is the most profitable one. We first note that that Proposition 1 immediately implies the following result.

COROLLARY 2. *The following conditions are necessary for technology k to be inducible:*

$$\Delta_{ki}(t_{ki}^{max}) \geq 0 \quad \textit{for all } i = 1, \ldots, k-1. \tag{A9}$$

However, it is not hard to demonstrate that the above conditions are necessary but not sufficient for inducibility.

To rigorously define the inducibility region for technology $j \in \{1,\ldots,k\}$ we proceed as follows. From Equation (9), for $i < j$ we have $t_{ij}^{crit} \leq t_{ji}^{crit}$ and technology j is at least as profitable as i iff $t \in [t_{ij}^{crit}, t_{ji}^{crit}]$. On the other hand, if $j < i$, then $t_{ji}^{crit} \leq t_{ij}^{crit}$ and j is at least as profitable as i iff $t \in [0, t_{ji}^{crit}] \cup [t_{ij}^{crit}, t_j^{lim}]$. Since technology j is cleaner than $i < j$ and dirtier than $i > j$, if we intersect the corresponding intervals and designate them with indices (C) and (D) (depending on whether j is cleaner or dirtier, respectively), we obtain the following result.

PROPOSITION 12. *The inducibility region of technology $j \in 1,\ldots,k$ is given by*

$$([0, t_{j*}^D] \cup [t_{*j}^D, t^{lim}]) \cap [t_{*j}^C, t_{j*}^C],$$

where $t_{*j}^C = \max_{i<j} t_{ij}^{crit}, t_{j*}^C = \min_{i<j} t_{ji}^{crit}, t_{j*}^D = \min_{i>j} t_{ji}^{crit}$, $t_{*j}^D = \max_{i>j} t_{ij}^{crit}$.

Note that by the preceding result, the inducibility region (if it exists) of some technology j consists of one or two closed intervals. We now turn our attention to the conditions for inducibility of the cleanest available technology k.

PROPOSITION 13. *Technology k is inducible iff $t_{*k}^C \leq t_{k*}^C$ and $t_{*k}^C < t_k^{lim}$.*

PROOF. Technology k is inducible iff $t_{*k}^C \leq t_{k*}^C$ and $t_{*k}^C < t_k^{lim}$. From the previous proposition and since k is the cleanest technology (i.e., has the highest index), it follows that the inducibility region of k is given by $[t_{*k}^C, t_{k*}^C]$. This regions is non-empty whenever the conditions in the proposition hold. □

We note that in order to check the condition for inducibility in the previous result, one has to solve k non-linear equations $\Delta_{ik}(t) = 0$ for $i = 1,\ldots,k-1$. Moreover, even if the condition in Proposition 13 holds but the interval $[t_{*k}^C, t_{k*}^C]$ is small, technology k may not be practically inducible due to uncertainties about model parameters. Corollary 1, however, extends directly to the k-technology case: the width of the inducibility region $t_{k*}^C - t_{*k}^C$ is non-decreasing in Θ_k suggesting that subsidizing the fixed cost of the cleanest technology may be a good strategy for expanding the inducibility region. The issue of subsidizing fixed costs is further considered in the next section.

Appendix D. Welfare Maximization with 3+ Technologies

Recall from Proposition 12 that the inducibility region for each technology consists of at most two disjoint intervals. Since for every t some technology must be induced, these regions partition the interval $[0, t^{lim}]$ into at most $2k - 1$ subintervals (since for technology k the inducibility region consists of a single interval). Thus, there exist $M \leq 2k$ *breakpoints* $t_{(m)}^B, m = 1,\ldots,M$

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designating the boundaries of these subintervals such that

$$t^B_{(1)} = 0, t^B_{(M)} = t^{lim}, \quad \text{and } t^B_{(m)} < t^B_{(m+1)} \quad \text{for } m = 1, \ldots, M-1.$$

Of course each of these breakpoints $t^B_{(m)}$ (for $1 < m < M$) is equal to a critical value t^{crit}_{ij} for some technologies i,j with i being the optimal technology for $t \in [t^B_{(m-1)}, t^B_{(m)}]$ (i.e., $i^*(t) = i$ in this interval) and j being optimal for $t \in [t^B_{(m)}, t^B_{(m+1)}]$.

Note that while the firm's profit is the same whether a given breakpoint is approached from the left or the right (this follows by the definition of the critical points), the same is not true for the optimal production quantity, and hence for emissions and other components of the welfare function $Q(t)$. Indeed, letting $x(t) \equiv x_{i^*(t)}(t)$ denote the optimal production quantity corresponding to tax rate t, we see that for $t^B_{(m)} = t^{crit}_{ij}$ the limit of $x(t)$ as $t \uparrow t^B_{(m)}$ from the left is $x_i(t^B_{(m)}) \equiv x_i(t^{crit}_{ij})$, while the limit of $x(t)$ as $t \downarrow t^B_{(m)}$ from the right is $x_j(t^B_{(m)}) \equiv x_j(t^{crit}_{ij})$, and from Equation (5) we know that, in general, $x_i(t^{crit}_{ij}) \neq x_j(t^{crit}_{ij})$.

We will use the notation $t^{B-}_{(m)}$ and $t^{B+}_{(m)}$ to designate left and right limits at this breakpoint; that is, $x(t^{B-}_{(m)}) = x_i(t^B_{(m)})$ and $x(t^{B+}_{(m)}) = x_j(t^B_{(m)})$. The revenue function $Q(t)$ is continuous on each interval $[t^{B+}_{(m)}, t^{B-}_{(m+1)}]$ since it coincides with some $Q_j(t)$ function on this interval, but will, in general, have discontinuities at each breakpoint. In fact, since the firm is indifferent between the two production quantities at the breakpoint because they earn the same profit, the welfare function is not well defined at the breakpoints. This has important practical implications, since both the firm and the regulator are very sensitive to small perturbations of the tax rate around each breakpoint: the welfare changes abruptly as the tax rate is changed from just below the breakpoint to just above it. In practice we may assume that there exists some small value $\mu > 0$ such that $t^{B-}_{(m)} = t^B_{(m)} - \mu$ and $t^{B+}_{(m)} = t^B_{(m)} + \mu$. With this convention we will regard the right and left limits at each breakpoint as two different tax rates.

Let $j(m)$ designate the induced technology for $t \in [t^B_{(m)}, t^B_{(m+1)}]$. Since $Q(t) = Q_{j(m)}(t)$ on this interval, the maximum value within this interval occurs either at one of the endpoints or at $t^{opt}_{j(m)}$ defined by Equation (16), if this value falls within the interval. We say that $t^{opt}_{j(m)}$ is *feasible* if it falls in the interior of the interval.

Let $M^f = \{m \in \{1, \ldots, M-1\} \mid t^{opt}_{j(m)} \text{ is feasible}\}$ (this set may be empty). The preceding discussion leads to the following result.

PROPOSITION 14 *Define the following discrete set of tax rates:*

$$T = \left\{ t^{B+}_{(m)}, t^{B-}_{(m)} \mid m \in \{2, \ldots, M-1\} - M^f \right\} \bigcup \left\{ t^{opt}_{j(m)} | m \in M^f \right\}.$$

Then the welfare-maximizing (socially optimal) tax rate t^{opt} can be found in T.

To summarize, the following algorithm can be used to compute the socially optimal tax rate:

1. Compute inducibility region for each technology using Proposition 12. The boundaries of these regions yield the set of breakpoints and their number M.
2. Use Equation (16) to compute t^{opt}_j for each technology j and identify whether this value is feasible (by checking whether it falls within the inducibility region of j).
3. Form set T defined in Proposition 14 and evaluate $Q_{i^*(t)}(t)$ for each element of T. The element yielding the maximum value is the socially optimal tax rate t^{opt}.

Notes

[1]http://www.mckinseyquarterly.com/article_print.aspx?L2= 21&L3= 0&ar=2077.

[2]An implicit assumption in our study is that a variety of technologies are available to be purchased and installed, which we believe is reasonable given the rapidly maturing market for clean technologies, for example, see Americanventuremagazine.com (American venture magazine 2007), or check the US EPA Technology Fact Sheets (US Environmental Protection Agency n.d.a, n.d.b).

[3]Note that if the technologies under consideration have different useful lives, then Θ's should be updated to represent, for example, annualized cost (amortization).

[4]To be consistent with the cost structure, demand should reflect the same time horizon as the cost; if cost is annualized then $D(p)$ represents the annual demand; if cost is for the entire useful life of the technology, then the demand should also aggregate the demands for that period; and so on.

[5]As mentioned in the Introduction, another way to interpret ε is as a weight parameter in the regulator's bi-objective problem.

[6]In practice, subsidy is awarded before tax revenues are collected; hence, providing a subsidy incurs borrowing from other sources or externally. Our analysis assumes this cost away.

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